

BOX HW4

Minimal information for a node-centric de Bruijn graph

For a node-centric de Bruijn graph of order k , the minimal information needed is the set of k -mers, which are the vertices. Edges do not need to be stored, they can be recomputed at any moment.

To account for both strands, we store only the canonical form of each k -mer, and canonicalize every k -mer before a membership test. One entry thus represents both strands, which halves the memory needed.

In practice, we store a dictionary mapping each canonical k -mer to its number of occurrences rather than a plain set. The keys are still the vertices, so no information is lost, and the multiplicities are needed later. Python's `Counter`, a dictionary subclass designed for counting, is well suited here for counting occurrences.

Multiplicity of the k -mers

Fasta file	k -mers processed	Distinct k -mers
Perfect reads, forward	2100000	149 866
Perfect reads, both strands	2100000	149 869
0.1% errors	2100000	213 031
1% errors	2100000	702 101

The multiplicity distribution has two parts. The perfect datasets show a single bell centered around 13-14, and a thin tail beyond 30 corresponding to repeated regions of the genome. The erroneous datasets add a large peak at multiplicity 1, made of k -mers created by sequencing errors. At 1%, the bell is also shifted to the left.

Threshold t values

On the erroneous reads, most k -mers seen only once are sequencing errors: the multiplicity histogram shows a peak at 1, clearly separated from the bell of true k -mers (centered around 13-14). Using the graph of the perfect reads as ground truth, false positives (erroneous k -mers kept) drop from 552 239 at $t=1$ to 8 219 at $t=2$ and 20 at $t=3$ for the 1% dataset (63 163, 76 and 0 for 0.1%), while false negatives (true k -mers lost) grow by a factor of about 2 to 4.5 per step (113, 513 and 1 538 for $t=2, 3, 4$ at 1%).

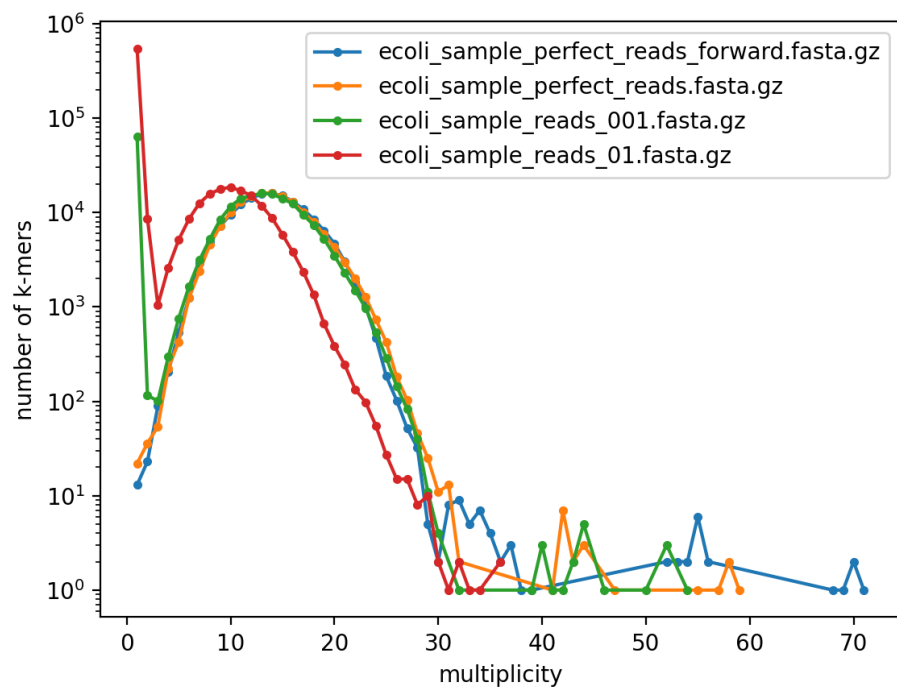


Figure 1: Multiplicity of k-mers for each files

I choose $t=3$, which minimizes false positives + false negatives on both datasets (56 for 0.1%, 533 for 1%). At $t=4$ all false positives are gone but three times more true k-mers are lost (1% dataset), whereas at $t=2$ thousands of errors remain. The 20 remaining false positives at $t=3$ (1% dataset) are likely errors repeated at least three times, which no threshold can separate from true k-mers. The best t also depends on the coverage.

Recovering unitig

Algorithm, some principles

- Two way search: The function builds the unitig by exploring the graph to the right and to the left
- Stopping at branches: It stops when a node has more than one successor, or if the successor has more than one predecessor
- Avoiding cycles: It stops if it finds a k-mer that was already visited, to prevent cycles

Results

Size (in bp) of the unitig containing the given 31-mer.

Dataset	$t = 1$	$t = 2$	$t = 3$	$t = 4$
Perfect reads, forward only	5 586	5 575	5 574	5 567
Perfect reads, both strands	5 588	5 581	3 243	3 243
0.1% errors	46	4 811	5 581	5 332
1% errors	31	757	3 095	2 243

On the perfect datasets, the unitig is nearly identical for both at $t=1$ and 2, which also checks the handling of both strands. It covers only about 4% of the 150 kbp region, so without errors it is most likely a repeat creating a fork (not verified). It decreases when t increases, because true k-mers of low multiplicity are removed, and each removal cuts the path.

On the datasets with errors, we see two effects. At low t , erroneous k-mers create forks that stop the unitig almost immediately (46 bp and 31 bp at $t=1$). When t increases, the false positives disappear and the unitig grows: 5 581 bp at 0.1% for $t=3$, identical to the perfect reads at $t = 2$! The size is therefore maximal around $t=3$, the threshold chosen. However, a single missing k-mer is enough to cut the unitig, so this size depends on the sample of reads. With 1% errors, about 27% of k-mers contain an error (against 3% at 0.1%), so the unitig stays shorter for every t than at 0.1% and never exceeds 3 095 bp.

Sum of the size of the recovered unitigs

Dataset	$t = 1$	$t = 2$	$t = 3$	$t = 4$
Perfect reads, forward	150 496	150 483	150 490	150 701
Perfect reads, both strands	150 499	150 477	150 531	150 687
0.1% errors	419 101	150 888	150 683	150 822
1% errors	2 233 421	194 095	152 526	155 321

On the perfect reads, the total size is about 150.5 kbp, whatever the threshold. At $t = 1$ the datasets with errors give much larger totals (419 kbp at 0.1%, 2.23 Mbp at 1%). This is not additional genome: erroneous k-mers form thousands of short unitigs, each adding its own overlap. The total drops at $t=2$ and is minimal at $t=3$ for both datasets, which confirms (again) the choice of Q7. It increases again at $t=4$ because true k-mers with low multiplicity are removed.